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# Railtrack Company Guidance Note

## Methodology for the Demonstration of Electrical Compatibility with Double Rail Reed Track Circuits on the DC Railway



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## IMPLEMENTATION

This code of practice is for the guidance of all persons who are responsible for the production, submission and review of EMC safety cases for Railtrack ESRP.

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# Contents

# 1. INTRODUCTION

## 1.1 Purpose

The purpose of this document is to provide a methodology to demonstrate electrical compatibility with 'RT' type double rail reed track circuits on the d.c. electrified railway on Railtrack controlled infrastructure. This is achieved by:

- providing a characterisation of the reed track circuit receiver;
- providing a characterisation of the infrastructure;
- providing a methodology to define the susceptibility of a given reed track circuit.

Combining these elements will enable a full characterisation of reed track circuits applicable to Railtrack controlled infrastructure, for both worst case intact and faulty infrastructure conditions. This characterisation will take the form of a permissible train current which can then be compared to the emission footprint of an item of rolling stock. This methodology will facilitate the demonstration of compatibility of rolling stock with reed track circuits, as part of a Route Acceptance Safety Case.

## 1.2 Scope

This document covers RT type reed track circuits installed in double rail mode on d.c. electrified Railtrack controlled infrastructure. This also encompasses the older type ‘R’ equipment when installed in double rail mode where these are not fitted with step up transformers.

The methodology considers both intact and faulty infrastructure.

## 1.3 Installation Types Covered by this Document

This document covers reed track circuits installed in double rail jointed mode on the d.c. electrified railway. It does not cover track circuits used in jointless mode, specific single rail installations that may be present within dual (a.c. and d.c.) immunised areas, or double rail installations in non d.c. electrified areas (used in tunnels etc.). Installations not specifically included in this document will require a specific assessment based on the configuration as installed. Installations on a.c. electrified lines are generally installed in single rail configuration and are therefore not covered in this document. These installations are covered in Ref. 2.2.6.

There may be instances of type ‘R’ reed track circuits that have been installed with step-up transformers fitted to the receiver, these track circuits are also excluded from the scope of this document.

The analysis here assumes that the d.c. electrification is installed as either a 'classic' or 'high power' third rail, 750V system (Ref. 2.2.2).

### 1.4 Equipment Types Covered by this Document

The equipment covered within this document is the type 'RT' reed track circuit equipment (Ref. 2.5.2), manufactured by GEC (now ALSTOM). This equipment has an adjustable track filter that affects the sensitivity of the receiver. For the analysis presented here the filter is assumed to be set at its most sensitive setting. At this setting the RT type reed receiver is more sensitive than the earlier type 'R' reed track circuit equipment. Due to this difference in susceptibility characteristic type 'R' reed track circuit equipment is encompassed within the analysis presented here.

This document considers both the rail connected receiver and the intermediate loop receiver configurations.

This document does not cover application specific receivers used in double rail track circuits on the a.c. railway nor does it cover more recently developed high power double rail reed track circuits that operate with higher rail to rail voltages than the type 'RT'.

## 2 SOURCE DOCUMENTS

### 2.1 Railway Group Standards

2.1.1 GK/RT0011, "Train Detection", Issue 3, August 2000.

2.1.2 GK/GN0611, "Train Detection", Issue 1, August 2000.

### 2.2 Railtrack Company Standards

2.2.1 RT/E/S/11752, "Train Detection", Issue 2, August 2001.

2.2.2 RT/E/S/21104, "Design & Installation of Electric Track Equipment for D.C. Electrified Lines", Issue 1, March 1998.

2.2.3 RT/E/PS/11765, "Impedance Bonds", Issue 1, December 2000.

2.2.4 RT/E/PS/11763, "Reed Type RT Track Circuits", Issue 1, December 2000.

2.2.5 RT/E/C/27010, "Compatibility between Electric Trains and the Electrification System", Issue 1, November 1997.

2.2.6 RT/E/C/50002, "Methodology for the Demonstration of Compliance with Single Rail Reed Track Circuits on the AC Railway", Issue 1.

### 2.3 Previous Safety Case Documentation

2.3.1 "Double Rail Reed Track Circuit Interference Limits – D.C. Routes", 10109.4106, Issue 3 [AEA / ALSTOM], June 1999.

2.3.2 “Determination of Permissible Rail/Wheel Interface Current Levels for Single Rail Reed Track Circuits on the AC Railway”, 120-dr0397, version b0, [AEAT Rail], May 1999.

### 2.4 Industry Data Initiative Documentation

2.4.1 1737001-C-TR-001 – “Strategy for the Development and Acceptance of ECPs”, Version 1.0, 18/11/99.

2.4.2 1737001-L-TR-001 – “Identification of Sources”, Version 0.1, 3/11/99.

2.4.3 SFCG07-DCReedTC – “Shortform Compatibility Guide, Volume 7 (Double Rail Type RT Reed Track Circuits (D.C. Electrified Areas))”, Version 1.0, 25/4/01.

2.4.4 ESID9079 – “Reed Receiver Test Report - Railtrack Industry Data Initiative”, AEAT Rail, , version 2, March 2000.

### 2.5 Other Documents

2.5.1 “Railway Safety - HM Chief Inspecting Officer of Railways’ Annual Report on the safety record of the railways in Great Britain during 1995/6” HSE Books.

2.5.2 “Type RT Reed Track Circuit Equipment Part 1 Standard Power Jointed Track Circuits” IB147 Part 1 Issue 4, [Alstom Signalling], October 1991.

2.5.3 "Determination of Rail Internal Impedance for Electric Traction System Simulations", RJ Hill & DC Carpenter, IEE Proceedings, Volume 138, No. 6, November 1991.

2.5.4 “Reed Track circuit parameters” fax from D. M. Poole to N Rambukwella [AEAT Rail], 12/5/93.

# 3 DESCRIPTION OF REED TRACK CIRCUITS

## 3.1 General Configuration for D.C. Electrified Railways

All track circuits in d.c. electric traction areas and for 800m beyond the limits of the traction area defined by the IRJ's (insulated rail joints) must be immune from d.c. traction interference (Ref. 2.2.1).

High traction return currents associated with low voltage d.c. traction systems require a very low impedance path back to the substation if voltage and power losses are to be kept within sensible limits for reliable train operation. This is achieved by using both running rails for traction return wherever possible, as well as cross bonding between parallel tracks, thereby reducing the overall return path impedance. Double rail track circuits allow both rails to be used for traction return currents. However in switch and crossing areas it is not normally possible to bond the track in double rail form and therefore single rail track circuits have to be installed.

In order to keep adjacent double rail track circuits electrically separate, IRJs (insulated rail joints) are used in both running rails to form jointed track circuits. So that the traction currents can flow past the insulated rail joints impedance bonds are required at both ends of the track circuit. Cross bonding to parallel tracks must be achieved without providing a substantial connection from one track circuit feed to another track circuit receiver.

An operational explanation of the use of impedance bonds to provide a low impedance path past the IRJs and avoid cross bonding providing connections from one track circuit feed to another is given in Ref. 2.2.3.

Provided that each rail is perfectly bonded and of equal resistance, the traction current will be equally divided and the impedance bonds will be perfectly balanced with no direct current flux in the cores. Unbalanced traction return currents may result from defective bonding but can also occur on long curves where the low rail is shorter and therefore offers less resistance. The effect can be particularly aggravated by the presence of the third rail.

## 3.2 Track Circuit Equipment

The type RT reed track circuit is a single audio frequency track circuit, designed and manufactured by GEC (now ALSTOM) that is normally installed in double rail mode on the d.c. railway (Ref. 2.5.2). The frequencies of operation (363Hz to 423Hz) and respective channel numbers (f211 to f221) are detailed in table 1 below:

| Channel number | Centre Frequency (Hz) |
|---|---|
| | |
| f211 | 363 |
| f212 | 366 |
| f213 | 369 |
| f214 | 372 |
| f215 | 375 |
| f216 | 378 |
| f217 | 381 |
| f218 | 384 |
| f219 | 408 |
| f220 | 417 |
| f221 | 423 |

Table 1: RT Reed Track Circuit Frequencies

On d.c. electrified lines, reed track circuits are usually restricted to the eight channels f211 to f218 (Ref. 2.2.4). However, this is not universally the case and it should be assumed that the higher frequency channels f219, f220 and f221 may be used, unless this is supported by route specific infrastructure data.

The type RT reed track circuit equipment is a development of the type R reed remote control equipment and the earlier type 'R' reed track circuit. It employs circuits which are tuned to a very high Q factor by means of the electromechanical tuning fork principle, having a pair of vibrating reeds which are used to produce a band pass filter which has a narrow bandwidth and a very stable centre frequency. For track circuit equipment, the frequencies are chosen to lie between harmonics of the mains power supply frequency and the standard sets of frequencies lie between the 7th and 8th harmonics and between the 8th and 9th harmonics, the narrow bandwidth allowing channel spacings down to 3 Hz.

A Reed track circuit consists of a transmitter which feeds audio frequency oscillations into the rail and a receiver at the relay end which is tuned to receive only that specific frequency. Immunity from sources of interference is achieved because the bandwidth of the receiver is so narrow.

The type RT Reed track circuit is capable of operating in jointless and jointed modes. However due to operating problems in the presence of traction bonding its jointless use is restricted to a few existing installations on non electrified lines. On d.c. electrified lines it is used in the jointed configuration, however, an intermediate receiver loop can be used in conjunction with the jointed track circuit to give a multisection track circuit. This is used, for instance, to indicate the start of the overlap for a signal when the track circuit end marks the end of the overlap. The shunting zone of the intermediate loop is variable such that the physical point at which a train is detected cannot be fully predicted. Intermediate loop receiver use is therefore restricted to lower criticality functions and is not a primary train detection facility.

The receiver loop is formed simply with the use of a loop of cable mounted in the four foot. The signal is detected in the rails by means of this loop and is fed via an attenuator filter and the receiver amplifier/filter to the track relay in the same manner as the end receiver. The loop acts as a step-up transformer, the running rails providing the primary turn and a 36 turn receiver loop forming the secondary. When there is no train shunt present, the normal track circuit signal current passes by the loop in one rail, then via the ballast and jointed receiver to return again past the loop in the other rail, thus inducing a voltage of that signal in the loop. Hence no output will result when a train shunts the track circuit on the Tx side of the loop. Traction return current (if in balance) will pass by the loop equally and in the same direction in each rail. Voltages induced in the loop will thus be cancelled out. If traction current is significantly out of balance then voltages could be induced in the loop which may cause the receiver equipment to malfunction.

Figures 1, 2 and 3 below show the basic arrangements of jointed reed track circuits (feed and receiver ends) and also a block schematic of an intermediate loop.

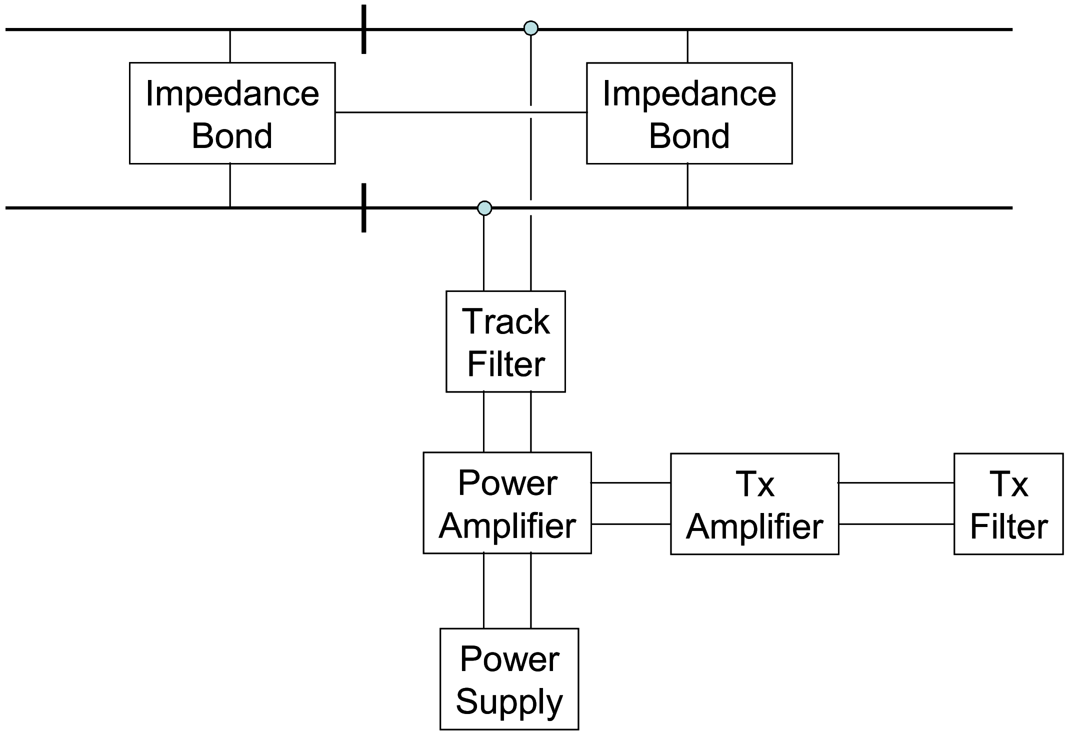


Figure 1: Block Schematic of Reed jointed Track Circuit (Feed End)

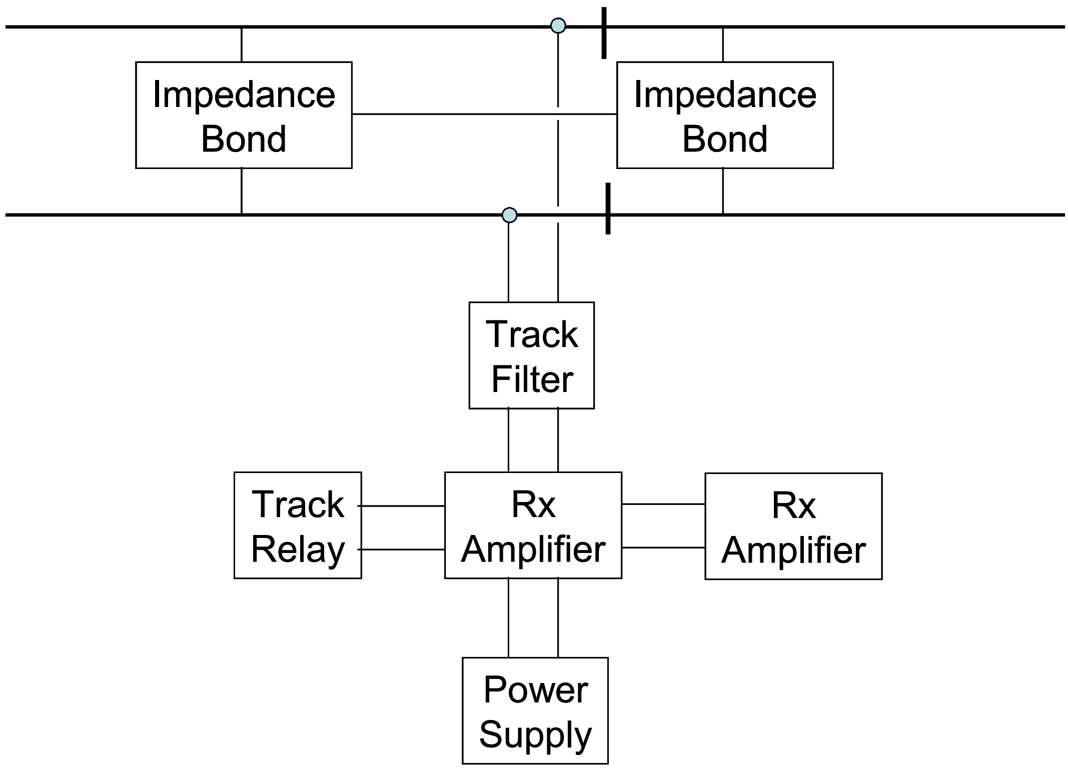


Figure 2: Block Schematic of Reed jointed Track Circuit (Receiver End)

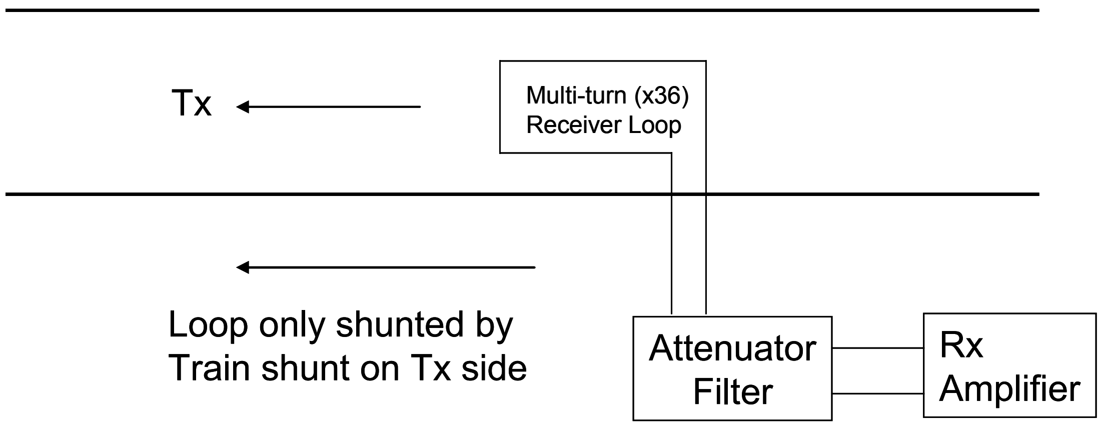


Figure 3: Block Schematic of an Intermediate Loop Receiver within a Reed Track Circuit

# 4 SUSCEPTIBILITY ANALYSIS

To determine the susceptibility of reed track circuits to electrical interference on the 750 VDC railway two pieces of existing work are employed.

- The track circuit receiver is characterized in terms of its response to electrical interference. This work is reported in Ref. 2.2.6.
- The infrastructure is characterized by way of a mathematical model of the railway which includes all of the conductors, cross bonds and impedance bonds. This work is reported in ref. 2.3.1.

## 4.1 Receiver Analysis

The reed receivers are the same whether they are used on the a.c. or d.c. railways. An extensive analysis into the susceptibility of reed receivers to electrical interference has been carried out in Ref. 2.2.6. This forms part of the work undertaken to determine the maximum permissible levels of reed frequency interference that can be tolerated on the a.c. railway as part of the IDI. Since this work is already documented and approved it is not necessary to repeat it here save for the results of that analysis.

Ref. 2.2.6 presents two sets of results and these are as follows:

- the maximum permissible reed filter voltage against time;
- the maximum permissible track filter primary current against time.

The results were obtained from testing and are summarized in figs. 4 and 5 below for the reed filter voltage and figs. 6 and 7 for the reed filter primary current.

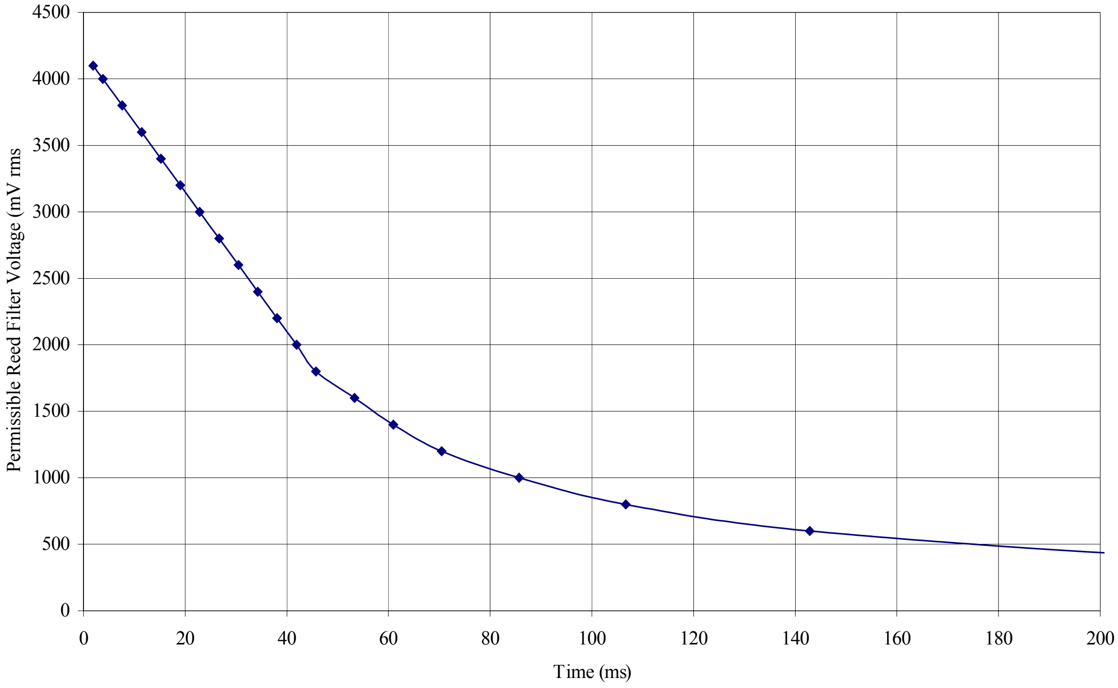


Figure 4 - Permissible reed filter voltage vs. time <200ms

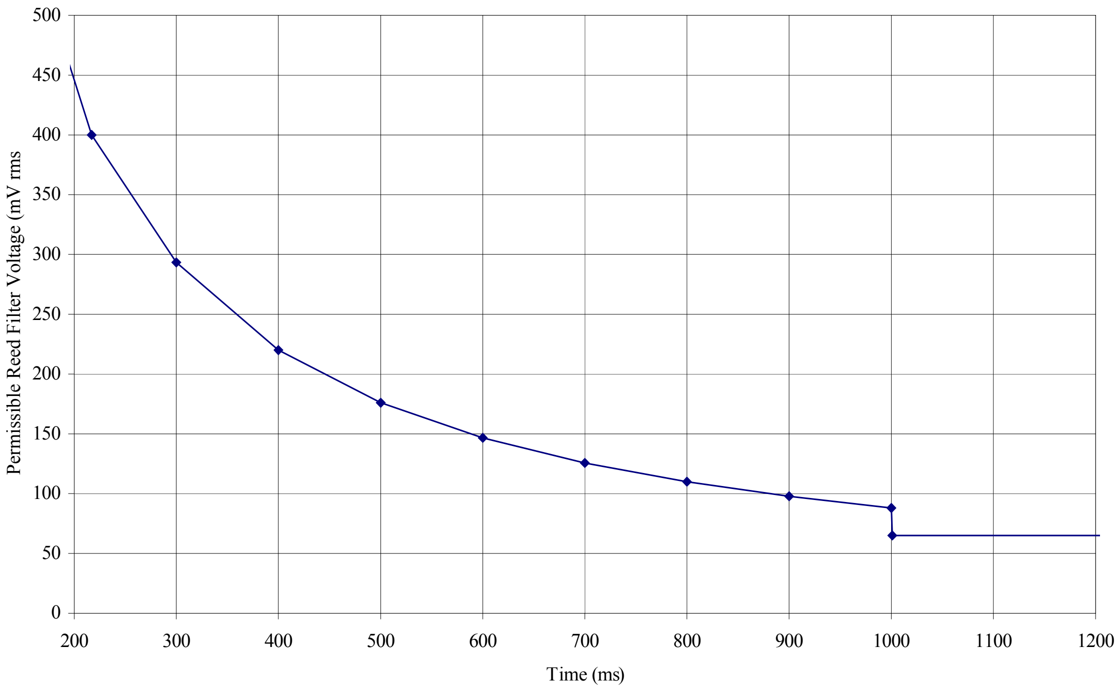


Figure 5 - Permissible reed filter voltage vs. time >200ms

The curve is horizontal at durations greater than 1 second since this voltage level is below that at which the receiver would 'drop'. This level is 65 mV rms.

The permissible reed filter primary current is obtained simply by applying the scaling factor of 1.6 mA/mV of the track filter (Ref. 2.2.6). This current is shown in figures 6 and 7. Again the curve is horizontal at durations greater than 1 second (steady state) the level being 104 mA rms and derived simply by multiplying the corresponding reed filter voltage of 65 mV by 1.6 mA/mV.

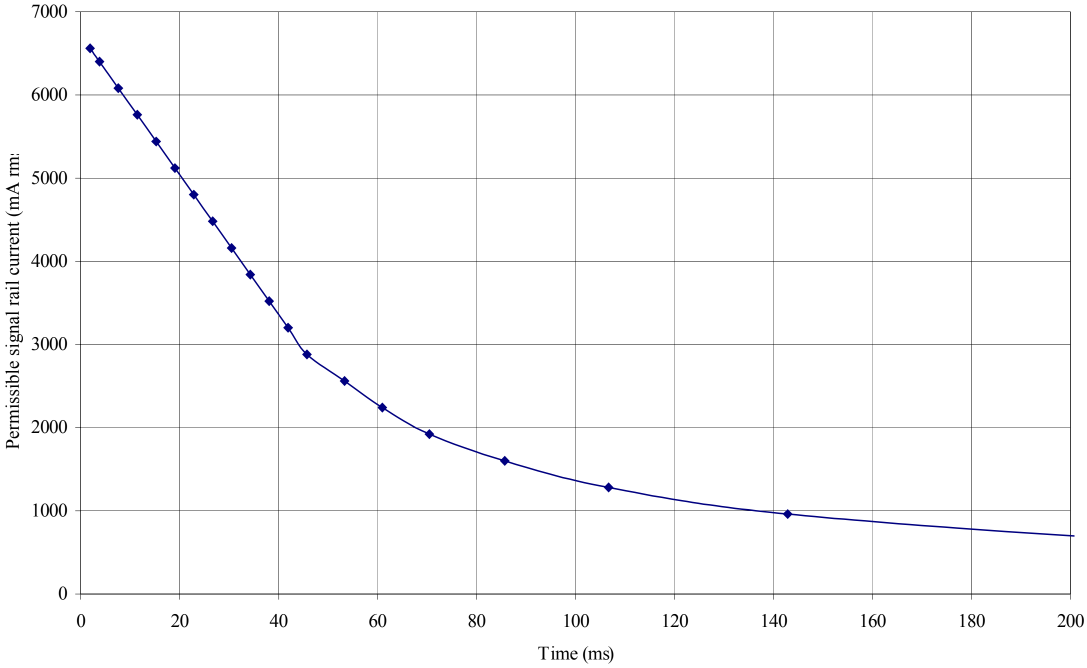


Figure 6 - Permissible track filter primary current vs. time <200ms

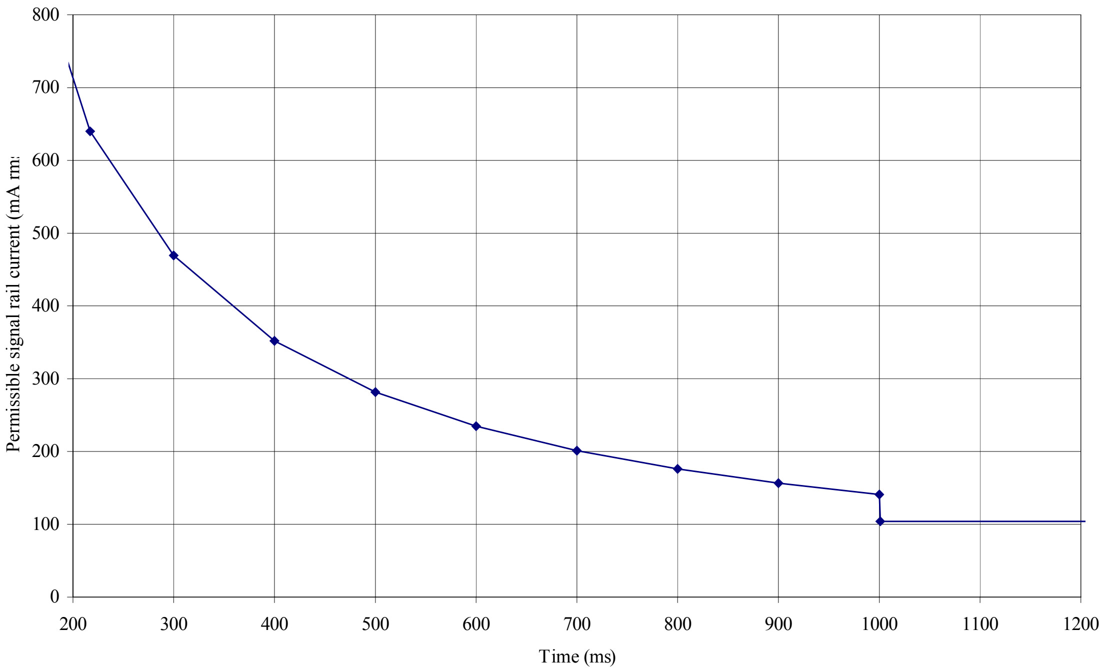


Figure 7 - Permissible track filter primary current vs. time >200ms

There are two plots for the permissible receiver voltage and two plots for the permissible track filter primary current. In each case one of the plots covers time intervals from zero to 200 ms and the other from 200 ms to what is considered to be steady state.

It should be noted that there is a step change at 1 second. This is because for the purpose of the analysis presented interference persisting for greater than 1 second (generally considered to be steady state) the track circuit receiver's drop away voltage was used whereas for less than 1 second it was the pick up voltage that was used (Ref. 2.2.6).

## 4.2 Infrastructure Characterisation

Ref. 2.3.1 presents an analytical investigation into the maximum permissible interference for trains in d.c. electrified areas fitted with double rail reed track circuits. This is achieved by characterising the 750 VDC third rail infrastructure system and considered three separate conditions:

- wrong side failure caused by interference current with all rails intact;
- wrong side failure caused by interference current with a broken traction return rail or impedance bond connection;
- wrong side failure caused by a train on an adjacent track producing axle to axle voltage.

Ref. 2.3.1 included an analysis of both the infrastructure and the track circuit (transmitter and receiver). The characterisation of the track circuit receiver has been superseded by that presented in Ref. 2.2.6; therefore for the purposes of this report only the infrastructure analysis from Ref. 2.3.1 is used.

For the intact infrastructure condition, a wrong side failure may occur if a longitudinal voltage is produced by current imbalance in the running rails.

The infrastructure analysis is that of an impedance network formed by the conductor rails, running rails, impedance bonds, cross bonds and the reed track circuit equipment. The infrastructure model included in Ref. 2.3.1 is used to calculate the voltage at the receiver for a given amount of interference current from a train. The infrastructure condition as modelled is shown in figure 8 below which shows a train shunting an intermediate impedance bond.]

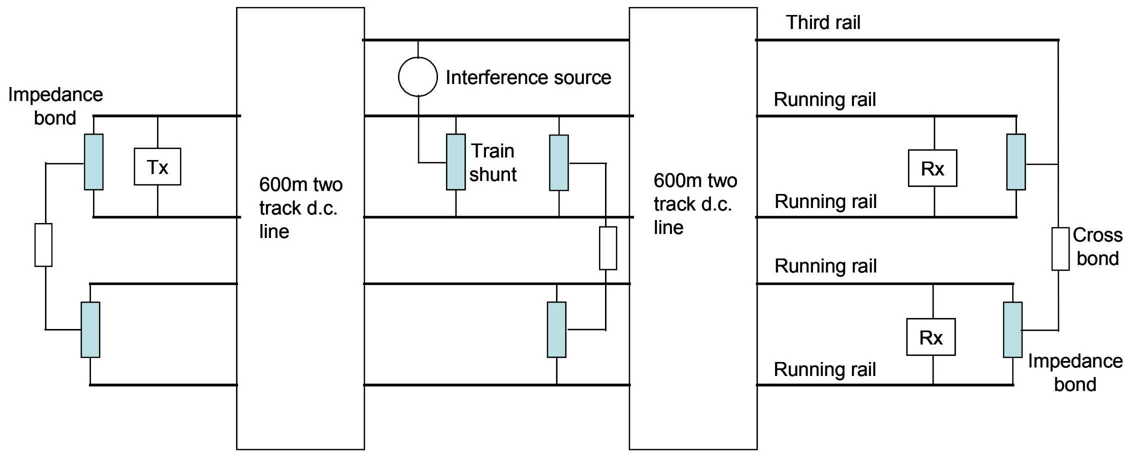


Figure 8 – Two Track Infrastructure Model.

For the failed infrastructure condition rail breaks (crosses numbered 1 to 10) are inserted as shown in figure 9 below.

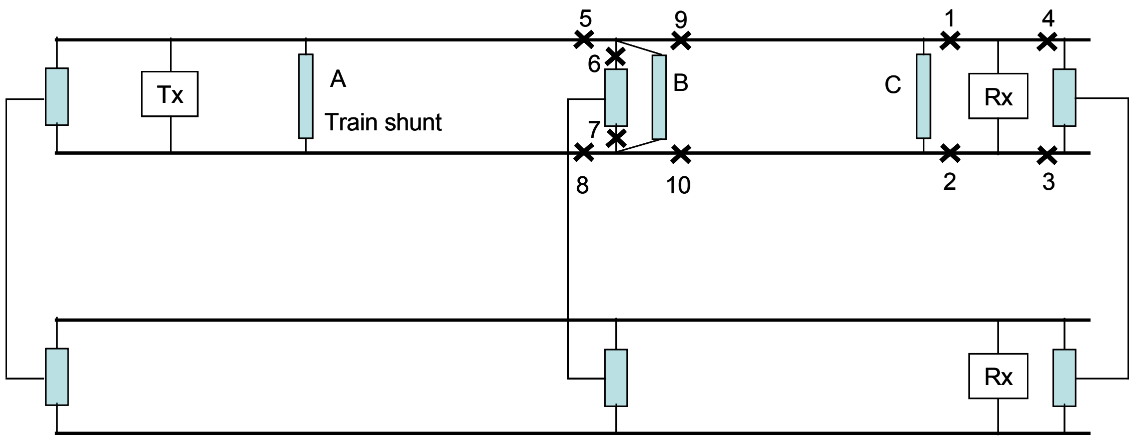


Figure 9 – Position of Rail Break and Train

For each of the rail breaks, the train was placed in the worst case position (i.e. the interference source) labelled A, B or C.

### 4.2.1 Intact Infrastructure

The intact infrastructure model based on the configuration shown in figure 8 is used in Ref. 2.3.1 to calculate the rail to rail voltage across the receiver that results from interference from a train. The calculations are based on a normalised interference current source of 1A, and since the model is linear, gives the scaling factor that can be applied to known levels of reed frequency interference produced by electric traction. Single rail configurations are modelled in Ref. 2.3.1 by increasing the resistor values representing the cross bonds to a high value and removing the intermediate impedance bond so that no current can pass.

From Ref. 2.3.1, the rail to rail voltage developed across the receiver for single and double track intact infrastructure resulting from a normalised 1A interference source is summarised in table 2 below. Included in table 2 is also the receiver filter current that would flow given the rail to rail voltage developed across it (given that the minimum receiver filter input impedance is 2.8Ω as stated in Ref. 2.2.6).

| Number of Tracks | Rail to Rail Voltage at Receiver | Receiver Filter Input Current |
|---|---|---|
| 1 | 439 mV | 157 mA |
| 2 | 449 mV | 160 mA |

Table 2 – Summary of Infrastructure Analysis for Intact Infrastructure

### 4.2.2 Failed Infrastructure

The failed infrastructure model based on the configuration shown in figure 9 is used in Ref. 2.3.1 to calculate the rail to rail voltage across the receiver that results from interference from a train. The calculations are based on a normalised interference current source of 1 A and, since the model is linear, gives the scaling factor that can be applied to known levels of reed frequency interference produced by electric traction.

From Ref. 2.3.1, the rail to rail voltage developed across the receiver for single and double track failed infrastructure resulting from a normalised 1A interference source is summarised in table 3 below. Included in table 3 is also the receiver filter current that would flow given the rail to rail voltage developed across it (given that the minimum receiver filter input impedance is 2.8Ω as stated in Ref. 2.2.6).

| Number of Tracks | Break Position (See Fig. 9) | Train Position (See Fig. 9) | Rail to Rail Voltage at Receiver (mV) | Receiver Filter Input Current (mA) |
|---|---|---|---|---|
| 1 | 1 | C | 1191 | 425 |
| 1 | 2 | C | 1191 | 425 |
| 1 | 3 | A | 678 | 242 |
| 1 | 4 | A | 1659 | 593 |
| 2 | 1 | C | 1116 | 399 |
| 2 | 2 | C | 1116 | 399 |
| 2 | 3 | A | 554 | 198 |
| 2 | 4 | A | 730 | 261 |
| 2 | 5 | B | 355 | 127 |
| 2 | 6 | A | 672 | 240 |
| 2 | 7 | A | 445 | 159 |
| 2 | 8 | A | 444 | 159 |
| 2 | 9 | B | 560 | 200 |
| 2 | 10 | B | 808 | 289 |

Table 3 – Summary of Infrastructure Analysis for Failed Infrastructure

### 4.2.3 Axle to Axle Voltage Condition

See section 4.3.4 below.

## 4.3 Track Circuit Susceptibility

### 4.3.1 Line Current Limits (End Receivers)

For the three infrastructure conditions considered, the receiver current values quoted in tables 1 to 3 above is that that would result given a normalised 1 amp interference source.

The infrastructure models are linear and as such the receiver current, as a percentage of the 1 amp source, is a transfer function. This transfer function, $\psi$, can be used in conjunction with the maximum permissible receiver current defined in Ref. 2.2.6 to determine the maximum permissible interference current that can flow in the third rail 750 VDC system.

The maximum permissible interference current is therefore given by:

$I_{rs} = I_{tds}/\psi$

Where:

$I_{rs}$ = The interference current produced by rolling stock.
$I_{tds}$ = The interference current flowing in the signal rail or primary of the track filter.
$\psi$ = The transfer function.

This equation can be applied to the curves shown in figures 6 and 7 to give the permissible interference current for reed frequency interference current of any duration.

This maximum train interference current can than be apportioned down to account for the effects of background interference that will also be present in the system from other sources. This includes transmitter breakthrough and trains within the same feeder section.

Using an apportionment factor of x3 for 'normal' infrastructure and x2 for 'faulty' infrastructure, the steady state limits (for continuous levels longer than 1000 ms) can be determined and are as detailed in table 4 below.

| No. Tracks | Condition | Permissible Rolling Stock Current |
|---|---|---|
| 1 | Infrastructure Normal | 221 mA |
| 1 | Infrastructure Faulty (Rail Break) | 122 mA |
| 1 | Infrastructure Faulty (Z-bond disconnection) | 88 mA |

| No. Tracks | Condition | Permissible Rolling Stock Current |
|---|---|---|
| 2 | Infrastructure Normal | 217 mA |
| 2 | Infrastructure Faulty (Rail Break) | 130 mA |
| 2 | Infrastructure Faulty (Z-bond disconnection) | 199 mA |

Note: The term 'Rail Break' here includes any series bonding disconnection.

Table 4 Steady State Rolling Stock Susceptibility Limits

### 4.3.2 Line Current Limits (Intermediate Loop Receivers)

For there to be any significant interference induced into an intermediate loop there must be significant imbalance in the return current carried by the running rails either side of the loop. The analysis presented in Appendix F considers the worst case scenario for interference from train current induced into a receiver loop: This is as follows:

- single track scenario;
- all return current in opposite direction to conductor rail current
- all return current in running rail remote from the conductor rail (e.g. due to a broken rail).

Under these worst case conditions, to develop the minimum threshold drop-away voltage of 65mV rms at the reed receiver filter input would require an interference current of 176mA rms.

An apportionment factor of x2 (applicable to faulty infrastructure is appropriate and this results in an applicable limit of 88mA. This is the same as the steady state limit for the end receiver.

### 4.3.3 Application to Non-Rectangular Shaped Transients

The transient event should be characterised as a square envelope having an amplitude of 90% of the peak value of the transient and a duration of the time taken for the transient to decay to 10% of its peak value.

If this characterisation sits below the derived permissible train current then no further consideration of this single transient event is required. However if the permissible train current curve is exceeded, alternative analysis would be required. For example the transfer function calculated could be used to scale a measured train current to enable this to be played into either a validated mathematical model of a reed receiver or a real receiver.

### 4.3.4 Axle to Axle or Longitudinal Voltage Limits

The train under consideration also poses a threat to track circuits installed on adjacent tracks in multi-track areas. Due to the flow of circulating currents within the train and the rails a voltage will be developed across the outermost axles of the train. This axle to axle voltage can act as a source to drive a current through the receiver of a track circuit on an adjacent track by virtue of the cross-bonding between the tracks (figure 10).

This may have an effect on the adjacent track circuit if the current caused to flow due to the axle to axle voltage is high enough to reach the drop away level of the track circuit. If this is the case the track circuit may fail to drop when occupied by a train.

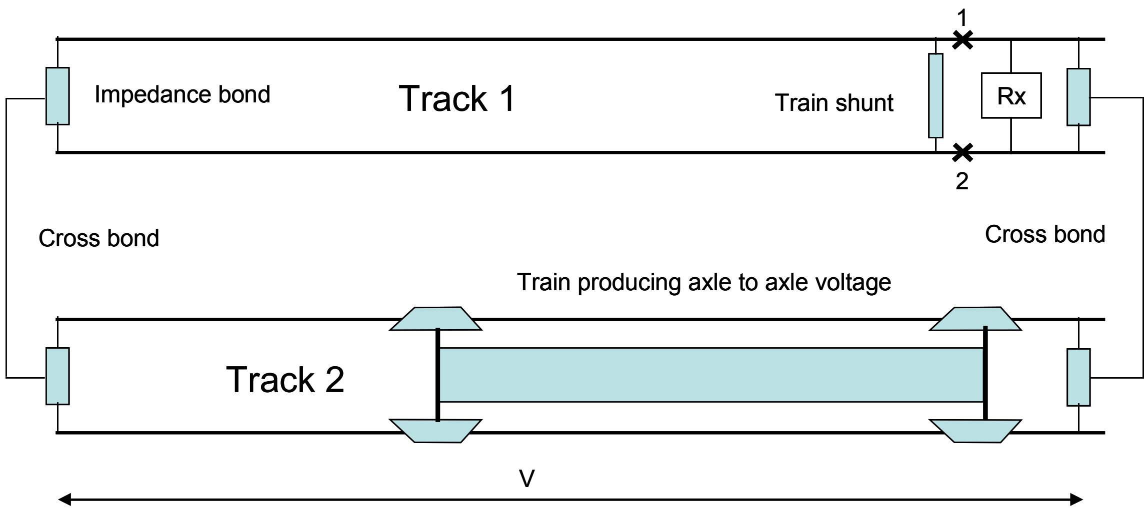


Figure 10 – Axle to Axle Voltage Condition

In practice it requires an infrastructure fault in the adjacent track circuit to present any significant proportion of the axle to axle voltage to the receiver track filter: pessimistically this proportion can reach 100% where there is no significant series rail impedance. From figure 7 the permissible signal rail current is 104 mA rms for 'steady state' conditions. The worst case condition for axle to axle voltage occurs when the train is the same length as the cross bond spacing. If the rail impedance is ignored, the axle to axle voltage that is required to drive 104 mA rms through the primary of the track filter (2.8Ω) would be 291 mV rms.

This value represents the absolute maximum voltage that can appear across the cross bonds and needs to be apportioned down to account for the presence of background and other sources of interference. If the same apportionment factor of x2 is used, this is reduced to 146 mV rms for the axle to axle voltage of the train under consideration.

# 5 BANDWIDTH

For the purpose of analysing train interference current, a half-power (-3dB) bandwidth of 0.5Hz may be assumed. As the correct operation of the reed equipment relies on a pair of vibrating reeds at both the transmitter and receiver end, then any significant drift in operating frequency of an individual reed rapidly results in a detected failure. The 0.5Hz bandwidth is therefore also considered sufficient to cover any drift in the operating frequency of the reed equipment.

However, in order to construct a robust argument, the bandwidth applied within the compatibility analysis should also consider the potential for particular interference components to drift in frequency from the particular equipment condition tested or analysed. It should also be noted that the spectrum analysis of extremes of mains frequency components may cause some spread into reed channels.

# 6 METHODOLOGY FOR DEMONSTRATION OF COMPATIBILITY

When configured as a double rail track circuit, the susceptibility level of a given track circuit depends upon the current imbalance between the rails, which for the limiting case of a broken rail or disconnected impedance bond, is not greatly affected by track circuit length.

The steady state limits given in table 3 (and the associated time/current characteristics by application of figures 6 & 7), can therefore be taken as representing a worst case, unless track circuits on a particular route are significantly longer than 1200m.

The longest track circuit on the route of interest may be determined by recourse to the RAR (Railtrack Asset Register) or other source. It is expected that this will be shorter than the 1200m track circuit analysed in Ref. 2.3.1.

It must be confirmed that the track circuits are installed in double rail mode only.

It must be confirmed that track circuits are of the RT type or if R type the receivers are not fitted with step-up transformers. Since the analysis is based on the most sensitive receiver setting there is no requirement to place any restrictions on future adjustment of reed track circuit receivers on the route except the fitting of step up transformers which must not be allowed.

The rail-break and impedance bond side lead failure figures and times at risk are indicative only and should be replaced by the latest available data and/or statements from Railtrack.

### 6.1 Normal Train - Normal Infrastructure

Once the longest track circuit on the route is determined, the procedure shown in section 3 can be followed to evaluate the worst case susceptibility for the track circuits on the route for the case of normal infrastructure. This is then reduced by the apportionment factor of x3 to determine the train current limits.

For the 1200m track circuit on double track examined in section 3 of this document the steady state limit is 217mA rms. This can then be compared to the train emission data. Under normal, including transient, operating conditions this limit must not be exceeded. It should be noted that the applicable figure of 221mA rms for the single track scenario is not significantly greater.

### 6.2 Faulty Train - Normal Infrastructure

From the failure analysis carried out the rate at which the train exceeds the curves calculated for normal infrastructure in section 3 should be determined. The frequency of occurrence, duration and magnitude of the exceedance should be taken into account when constructing mitigating arguments.

### 6.3 Normal Train - Faulty Infrastructure

The faulty infrastructure scenario as described in section 3 is used to determine a permissible time current curve for a given fault scenario. This can then be apportioned down to determine a train current limit. In the case of faulty infrastructure a reduced apportionment factor of x2 can be applied, due to the likelihood of immediate detection of the fault and limited time exposure where it is not immediately detected. This train current limit so calculated is 88 mA rms on single track sections and 130 mA rms on double or multiple track. This can then be compared to the train data.

In this case there may be specific normal transient events that may exceed the permissible current curve. These events should be quantified in terms of the rate at which they occur and the overall rate of potential WSFs is determined by combining them with track failure data.

This can be done in the following manner.

For the case of a broken rail occurring on a particular track circuit the total number of rail breaks per year is 680 (Ref. 2.5.1), per 20,000 track miles. Pessimistically a time at risk of 12 hours may be taken as the average time for a rail break to be discovered. The number of railbreaks and equivalent series bonding failures per year for the section of railway of interest must be confirmed by recourse to latest available figures or statements available from Railtrack. The probability of a track circuit having a broken running rail at any one time is:

Probability of track circuit having broken rail is:

(Ptcbr) = No. of breaks/track miles * TC length(miles)*(time at risk/No. of hours per year)

The predicted rate for a train which is standing on a track circuit exceeding the susceptibility level and the traction rail being broken is therefore:

Ptcbr * Exceedance rate (per hour)

This will then give the failure rate in exceedances per hour for normal train and failed infrastructure.

A similar approach can be taken with regard to the more specific impedance bond side lead failure, calculating the probability of a track circuit having a disconnected receiver end impedance bond side lead (Ptcbsl).

### 6.4 Faulty Train - Normal Infrastructure

In this case the train limits and the probability of a track circuit having a broken rail (Ptcbr), or a disconnected impedance bond side lead (Ptcbsl), are determined in the same way as in section 6.3. The exceedance rate for train failures is predicted by the failure analysis. This failure rate is then used in conjunction with the probability of a track circuit having a broken rail to determine the overall exceedance rate for this case.

### 6.5 Safety Targets

The overall exceedance rate shall be found by combining the rates calculated in sections 6.2, 6.3 and 6.4. This total exceedance rate shall be shown to be ALARP in the context of the safety requirements of the project.

### 6.6 Intermediate Loop Receivers

The analysis in Appendix F demonstrates that the worst case scenario for an intermediate loop receiver results in a susceptibility level the same as for a 1200m double rail track circuit. Therefore, provided that the interference requirements for end connected receivers are met, compatibility with intermediate loop receivers will be met.

### 6.7 Axle to Axle Voltages

The axle to axle voltage should be compliant with the limit of 146mV defined in section 4.3.4.

# APPENDIX A: PARAMETERS

Defined in Refs. 2.2.6 & 2.3.1 and summarised below.

All component parameters in the model were subject to sensitivity analysis by testing the effect of a +10% and –10% change in parameter value (10109.4106). This analysis demonstrated that no parameter change had an effect on the resultant transfer function that was more than linearly proportional. Therefore, given that all parameter values used were either pessimistic or typical maximum values, the transfer functions derived are assumed to be reasonably pessimistic.

| **Parameter:** | **Rail impedance** |
|---|---|
| Values: | 0.2mΩ/m: rail resistance<br>0.2μH/m: rail internal self inductance<br>0.48μH/m: conductor rail external self inductance<br>0.46μH/m: running rail external self inductance |
| Expected variation: | Overall impedance of running rail is:<br>1.56mΩ/m @ 363Hz,<br>1.77mΩ/m @ 423Hz<br>This is within 3% of being proportional to frequency, with the highest normally used frequency (384Hz) being used for the infrastructure analysis. |
| Sources: | • Refs. 2.3.1, 2.5.3 |

| **Parameters:** | **Railway layout dimensions** |
|---|---|
| Values: | 1.435m: distance between running rails<br>1.83m: distance between adjacent running rails of parallel tracks (6 feet)<br>0.076m: height of third rail above adjacent running rail<br>0.37m: horizontal distance between third rail and adjacent running rail |
| Expected variation: | Basic rail positioning requirements expected to be representative within at most 10%. |
| Sources: | • Refs. 2.2.5, 2.3.1 |

| **Parameters:** | **Impedance Bond Parameters (S2 type)** | |
|---|---|---|
| Values: | Tuning capacitor | 173nF - f218, 384Hz<br>182nF - f211, 363Hz |
| | Turns ratio | 1:56, (traction:aux coils) |
| | Loss resistance | 13Ω |
| | Winding resistance | 7Ω |
| | Magnetising inductance | 316.6µH |
| | Leakage reactance | 31.8mH |
| Expected variation: | Tolerance on manufactured/re-serviced S2 impedance bonds, expected to be within at most 10%. Sensitivity analysis has demonstrated that no parameter change has an effect on the resultant transfer function which is more than linearly proportional. | |
| Source: | • Ref. 2.3.1 | |

| **Parameter:** | **Reed receiver 'pick-up' criteria** |
|---|---|
| Value: | 88000 mV*ms, between 200ms and 1 second, based on further test results below 200ms, drop away criteria 65mV used for durations above 1 second. |
| Expected variation: | These limits are expected to be applicable for all RT type reed track circuit receivers. The 'pick-up' criteria is based on the most sensitive filter from a sample tested |
| Sources: | • Refs. 2.3.2, 2.4.4 |

| **Parameter:** | **Ratio of input current to output voltage for track filter** |
|---|---|
| Value: | 1.6 mA/mV |
| Expected variation: | This value is independent of the setting of the filter and is expected to be applicable for all RT type reed track circuit receivers |
| Source: | • Ref. 2.3.2 |

| **Parameter:** | **Track filter primary impedance at resonance** |
|---|---|
| Value: | 2.8Ω resistive |
| Expected variation: | This value is based on the track filter of an RT type reed receiver at its most sensitive setting and is expected to be applicable for all RT type reed track circuit receivers |
| Sources: | • Refs. 2.3.1, 2.5.4 |

## APPENDIX B: ASSUMPTIONS

A1 Configuration Details

A1.1 Track circuits are installed in double rail jointed mode, with limits defined by IRJs and traction return current normally balanced between the two running rails by the use of impedance bonds tuned to give a significant impedance at the track circuit frequency.

A1.2 All cross bonds between reed track circuits are connected via the centre tap of a tuned impedance bond, so as not to imbalance traction return currents.

A1.3 Track circuits are installed in double rail jointed mode, with limits defined by IRJs and traction return current normally balanced between the two running rails by the use of impedance bonds tuned to give a significant impedance at the track circuit frequency.

A2 Equipment Type

A2.1 Only type RT and type R receivers connected in double rail configuration, in accordance with the manufacturer's instructions (IB147, ref. 2.5.2) are considered.

A2.2 High power equipment is not included.

A2.3 Type R track circuit receivers are assumed not to have step up transformers fitted.

A3 General Infrastructure Assumptions

A3.1 The d.c. electrification is installed as either a 'classic' or 'high power' third rail, 750V system (Ref. 2.2.2).

A3.2 The track is assumed to be of standard gauge.

A3.3 CWR is assumed, as in d.c. electrified areas, traction return rails are bonded to give a similar resistance. Sensitivity analysis (Ref. 2.3.1) has shown that the derived susceptibility level is not at all sensitive to small changes in rail resistance.

A3.4 UIC 60 Rail is not included in the assessment as presented. The methodology given here is applicable but the parameters of this rail type at the frequency of interest will need to be determined and used in the appropriate calculation albeit that it is unlikely to have a significant effect.

A3.5 The presence of check rails will reduce the effective impedance of the rail to which they are attached. This is not included in this document, however the methodology provided allows for the effect of such differences to be evaluated.

A3.6 The sleepers in use are assumed to be concrete or wood, steel sleepers are not included in the analysis presented here.

A4 Types of Infrastructure Fault Considered

A4.1 The analysis presented considers only one infrastructure fault (an impedance bond side lead disconnection, cross-bond disconnection or a series bonding failure, including a rail break), occurring at any one time.

A4.2 A cross bond failure for the two track case is equivalent to the 'normal' infrastructure single track railway case, which has a susceptibility limit higher than any two track scenario.

A4.3 The broken rail scenario examined considers a broken rail at the worst position with regard to its position relative to receiver and train location. The length of the train is ignored, whilst in reality the length of the traction return path would be reduced by the train length leading to less current flow through the track filter.

A5 Analysis Assumptions

A5.1 A maximum length track circuit of 1200m has been modelled (Ref. 2.2.4).

A5.2 Transients are not repetitive (analysis considers response to single transients only, with no reset time being defined).

A5.3 Results from tests conducted on receiver equipment of one operating frequency can be applied to other operating frequencies.

A5.4 All of the traction current returns to source via the rails. Ballast resistance is assumed to be infinite: the effect of earth leakage from the conductor and running rails is ignored. The effect would tend to reduce the current returning to source via the running rails.

A5.5 The effects of inter-rail capacitance are ignored.

A5.6 The interference source was positioned to achieve the maximum effect on the track circuit receiver.

A5.7 The infrastructure fault was positioned to achieve the maximum effect on the track circuit receiver.

A5.8 The train is modelled as a point interference current source when calculating the permissible traction generated interference current.

A5.9 The track circuit current flow analysis was conducted for channel f218 (384Hz), which is the highest frequency normally used in d.c. electrified areas, with a channel f211 (363Hz) track circuit on the adjacent track.

A5.10 A maximum of 100% of axle to axle voltage can be presented across a receiver track filter.

A5.11 The transmitters of track circuits on adjacent tracks are assumed to be open circuits., although these use the same track filters as the receiver and will conduct some traction current.

A5.12 A cross-bond spacing of 600m was assumed for the double track model. Sensitivity analysis (Ref. 2.3.1) has shown that repositioning the intermediate impedance bond will not significantly change the analysis results.

A5.13 The intermediate receiver loop comprises 36 turns and is assumed to be of length 5m (Ref. 2.2.4).

A5.14 The intermediate receiver loop cable is installed in very close proximity to the running rail. Assuming an effective perimeter of 630mm, this gives a distance of 100mm between the centre of the running rail and the adjacent loop conductor.

A5.15 Whilst restrictions exist regarding the positioning of intermediate receiver loops in relation to the ends of jointed track circuits, these relate to signalling timing requirements and non-compliance with these will not affect the loop receiver's susceptibility to traction interference.

# APPENDIX C: HAZARDS

The following table lists the hazards identified that are relevant to the reed track circuit equipment.

| Hazard | Close out |
|---|---|
| Interference current in the traction return. | Covered by the assessment as presented in Refs. 2.2.6 & 2.3.1 and summarised in this document. |
| Axle to axle or longitudinal voltages. | Covered as presented in this document. |
| Direct induction into track mounted or trackside components | Direct induction into track circuit equipment has not been specifically considered in this assessment as the interference current levels required would be much higher than those derived for the conducted interference mode in this document. The same reasoning as for the a.c. railway case applies (Ref. 2.2.6). |
| Transient events. | Gapping transients and filter inrush transients have been specifically identified as containing significant reed frequency current. However, most transients may contain reed frequency interference therefore train testing must include all foreseeable transient conditions |
| Wheelset Impedance | The maximum wheelset impedance is not quoted in this document. See Ref. 2.1.1. |
| Vehicle geometry (overhang, wheelbase etc..) | This hazard is not covered by this document. See Ref. 2.1.1. |
| RSF due to excessive off channel current | This hazard is not covered by this document. |

# APPENDIX D: MITIGATING FACTORS

M1 Pessimisstic Assumptions

M1.1 The minimum track filter primary impedance: 2.8Ω resistive, has been applied in the analysis: representing selection of the highest sensitivity tapping. In practice, track circuits on a particular route may utilise different track filter tappings, which would reduce the receiver filter voltage for a given track filter primary current.

M1.2 The train is modelled as a point interference current source, with zero length. For multiple unit operation with a minimum consist length, the inclusion of the track length shunted by the train may be significant.

M1.3 The inclusion of earth leakage from the conductor and running rails tends to reduce the current returning to source via the running rails.

M1.4 The interference source was positioned to achieve the maximum effect on the track circuit receiver. The probability of transient interference occurring within the section of the track circuit which gives greatest susceptibility, could be significant in some circumstances.

M1.5 The maximum length track circuit of 1200m was used in the analysis. In practice, track circuits on a particular route may be significantly shorter and less susceptible.

M1.6 The train is assumed to run on at the transmitter end which may not always be the case.

M1.7 It has been assumed that 100% of the axle to axle voltage can be presented across a receiver track filter. This requires an infrastructure failure, and a short cross-bond spacing (just longer than the train).

M1.8 The transmitters of track circuits on adjacent tracks are assumed in the analysis to be open circuits., although these use the same track filters as the receiver and will conduct some traction current.

M1.9 Exceedance of permitted current curve may only be transient in nature, leading to a short duration, time limited WSF

M1.10 A higher permissible level would be applicable to the High Performance variant of reed RT track circuit.

M1.11 No TPR is assumed.

M1.12 It is assumed that there is no additional series impedance provided by the attenuator filter used with the intermediate receiver loop

M2 Assumptions/variations Covered by Apportionment factors

M2.1 Sensitivity analysis has demonstrated that no parameter change has an effect on the resultant transfer function that is more than linearly proportional. The expected normal variation of parameter values, such as track circuit receiver performance is in the order of +/-10%: this is small compared to the apportionment factors applied. Therefore, given that all parameter values used were either pessimistic or typical maximum values, the transfer functions derived are assumed to be reasonably pessimistic.

M3 Background/Coincident Sources of Interference Covered by Apportionment factors

M3.1 Other rolling stock.

M3.2 Transmitter breakthrough.

M3.3 To add constructively, the interference from other sources must be of the correct frequency due to the narrow band nature of the filter, and must have the correct phase relationship.

## APPENDIX E: SOURCES OF TECHNICAL WORK

The technical work contained in this document is taken from a number of sources:

- The Class 92 Safety Case, produced by AEA Technology Rail. ISA was WS Atkins Rail and Mott MacDonald.

The D.C. Juniper (Class 458 & 460) Safety Cases, produced by AEA Technology Rail and Alstom Transport. ISA was WS Atkins Rail.

# APPENDIX F: INTERMEDIATE LOOP RECEIVERS

The d.c. infrastructure model is shown in block form in figure 11 below. The detailed circuit model can be found in Ref. 2.3.1. This represents a double track section of railway with two parallel reed double rail track circuits of the maximum specified length of 1200m (Ref. 2.2.4).

The train is represented by a current source and associated shunt resistances.

The mutual impedances between traction system conductors (two running rails and conductor rail per track) are included in the model.

The impedances of the connecting bonds between the impedance bonds in each track can be set to open circuit to model the single track configuration. In multiple track configurations (i.e. more than two parallel tracks), the additional current sharing between parallel tracks reduces the proportion of traction return current that will flow in any particular track circuit, thereby reducing its susceptibility.

- **1.435m** distance between running rails (Ref. 2.2.5). Whilst this measurement is for the distance between inside running edges, analysis (Ref. 2.3.1) shows that the use of 1.505m to represent the distance between running rail centres has insignificant effect on the results.
- **1.83m** (6 feet) distance between adjacent running rails of parallel tracks.
- **0.076m** height of third rail above adjacent running rail (Ref. 2.2.5)
- **0.37m** horizontal distance between third rail and adjacent running rail (Ref. 2.2.5)
- **0.63m** perimeter of BS113A running rail (Ref. 2.5.3)
- **0.58m** perimeter of conductor rail (Ref. 2.5.3)
- **0.2mΩ/m** rail resistance (Ref. 2.3.1)
- **0.2μH/m** rail internal self inductance (Ref. 2.3.1)
- **0.48μH/m** conductor rail external self inductance (Ref. 2.3.1)
- **0.46μH/m** running rail external self inductance (Ref. 2.3.1)

Worst case analysis is for all traction current to be flowing passed the loop via the conductor rail and the opposite running rail.

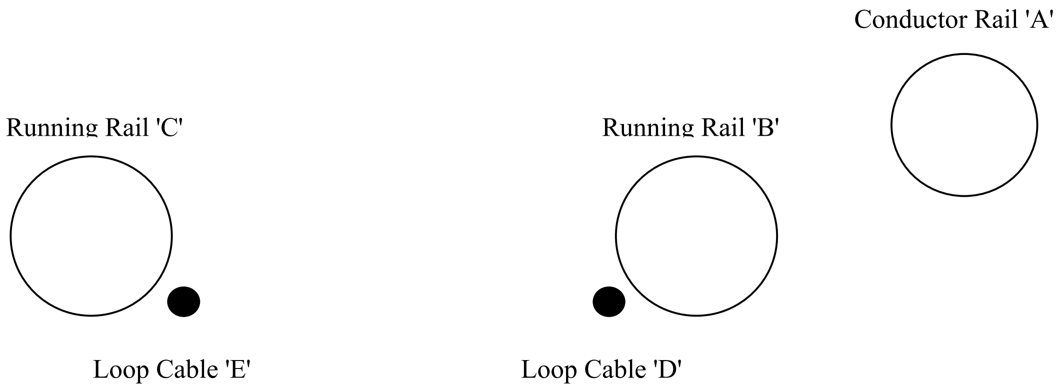


Figure 11 – Intermediate Loop: Induced Interference Mechanism

Considering the following conductors:

**A** conductor rail
**B** running rail adjacent to conductor rail
**C** running rail remote from conductor rail
**D** loop receiver cable adjacent to conductor rail
**E** loop receiver cable remote from conductor rail

The following distances can be derived:

**A-D** 478mm
**A-E** 1707mm
**B-D** 100mm
**B-E** 1335mm
**C-D** 1335mm
**C-E** 100mm

The basic formula for induced voltage in a third conductor due to current flow in a pair of conductors can be applied:

$V/I = \omega.\text{length}.2E\text{-}7.\text{Ln}(d_{\text{conductor 1}} / d_{\text{conductor2}})$

Where:

| | |
|---|---|
| V/I | Voltage per unit current (V/A) |
| length | Length of parallelism (m) |
| Ln | Natural logarithm |
| $d_{\text{conductor 1}}$ | Distance between conductor 1 and conductor 3 |
| $d_{\text{conductor 2}}$ | Distance between conductor 2 and conductor 3 |

When applied to the loop receiver scenario, the worst case (highest loop receiver voltage per unit current) is given by the situation where all the return current is concentrated in the running rail remote from the conductor rail (e.g. due to a broken rail). The net voltage developed across the terminals of 36 turn loop can therefore be calculated as follows:

$$V/I = 36.2.\pi.423.5.2E\text{-}7.[\mathrm{Ln}(d_{A\text{-}D}/d_{C\text{-}D}) - \mathrm{Ln}(d_{A\text{-}E}/d_{C\text{-}E})]$$
$$= 0.370\ V/A$$

Therefore, for a threshold drop-away voltage of 65mV rms at the reed receiver filter input, this equates to a train current of 176mA rms. Applying an apportionment factor of x2 for faulty infrastructure, this results in an applicable limit of 88mA, which is the same as the steady state limit for the end receiver.

RAILTRACK

**BRIEFING NOTE**

**RT/E/C/50001 to RT/E/C/50019** **Issue 1**
**RT/E/G/50003; RT/E/G/00014 and RT/E/G/50016** **Issue 1**

**Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure**

**Issue Date: February 2003**

**Background / Synopsis**

This suite of Standards (Codes of Practice and Guidance Notes) is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

**Key Changes / Compliance**

This is a new suite of documents. As the documents are Codes of Practice or Guidance Notes, compliance is not mandatory.

**Implementation**

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

**Of Interest To:**

Authors of Route Acceptance Safety Cases.

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